

Virad Munir

Lahore, Pakistan

virad.munir1189@gmail.com | <http://linkedin.com/in/viradmunir12> | <https://github.com/ViradMunir> | +92 317 6194972

Curious about building AI-driven and software-intensive systems using Python and C++. Skilled in machine learning, computer vision, robotics, web development, with hands-on experience deploying intelligent systems on edge hardware.

EDUCATION

Lahore University of Management Sciences

Sept 2022 - June 2026

BS Electrical Engineering (CGPA: 3.43)

Minor in Computer Science

Relevant Courses: Data Structures, Databases, Software Engineering, Foundations of AI and Machine Learning, Deep Learning, Principles and Techniques of Data Science, Junior Design Studio – Robotics

Beaconhouse School System

Sept 2018 – May 2022

A-Level – 3A*, O-Level – 6A* 2A

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, ROS

AI Expertise: YOLO, CNNs, Computer Vision, Object Detection, Deep Learning, RAG

Frontend: JavaScript, React.js

Backend: REST, PostgreSQL

Tools: Git, Linux, Jupyter, Google Colab, Kaggle, Roboflow, CUDA, ChatGPT

ACADEMIC PROJECTS

AI-Enabled Unmanned Ground Vehicle for Under-Canopy Crop Scouting (Final Year Project)

- Developed low-cost, autonomous under-canopy agricultural rover using computer vision, IMU-based navigation, and depth sensing for GPS-free crop-row traversal.
- Trained and evaluated a YOLOv26 object detection model on a custom wheat dataset of 1,163 annotated images for detection of wheat heads, tillers, and stripe rust, achieving 82% mAP @ 50 across all classes with real-time inference at 10 FPS on Raspberry Pi 5.
- Implemented real-time obstacle avoidance and row-end turning pipelines using RGB-D perception, OpenCV, and Python on a Raspberry Pi 5 and ESP32-S3 embedded system.
- Conducted field validation of autonomous navigation and perception systems in real agricultural environments for early crop monitoring.

Enterprise Document Assistant (RAG)

- Built a Retrieval-Augmented Generation (RAG) application enabling natural-language question answering over multiple PDF documents.
- Implemented a document ingestion pipeline using PyPDFLoader, recursive text chunking, sentence-transformer embeddings, and ChromaDB for semantic retrieval.
- Integrated the Groq Llama 3.1 API with a Streamlit interface to generate responses grounded in retrieved document text.
- Added document source attribution by displaying the originating PDF and page number for retrieved answers.

Bitcoin Random Walk vs LSTM (Cryptocurrency Price Analysis)

- Trained an LSTM-based time-series model on historical Bitcoin data using engineered financial features and a 30-day lookback window to investigate short-term price prediction.
- Implemented data scaling, temporal train–test splitting, and sequence generation to prepare Bitcoin time-series data for deep-learning experiments.
- Visualized training and validation loss alongside predicted-versus-actual prices to evaluate model convergence and forecast behavior across different time scales.

SmartCart – E-Commerce Platform (Team Software Engineering Project)

- Developed backend services using Next.js API routes and PostgreSQL, implementing RESTful APIs for authentication, cart management, order processing, and database operations.
- Integrated AI-powered review summarization and chatbot modules into platform workflows.

Autonomous Corridor Mapping and Delivery Using DJI RoboMaster & Hector SLAM

- Built a 2D indoor mapping system using ROS and Hector SLAM.
- Implemented autonomous navigation, object delivery, and dynamic obstacle avoidance.

HONOURS & AWARDS

- Merit Scholarship 100% – A-Level Programme
- Dean's List of Honors – LUMS
- Gold Medalist – A-Level Programme
- Gold Medalist – O-Level Programme